Unconditional Cash Transfers and Youth Labor Market Trajectories: Evidence from South Korea *

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Abstract

This paper examines how unconditional income support affects labor supply decisions among financially dependent young adults. We study a universal cash transfer program in Seongnam, South Korea, which provided 1 million KRW (about 19.1% of annual earnings) in quarterly payments to all 24-year-olds for one year. Using administrative panel data and a difference-in-differences design, we compare eligible youth in Seongnam with those in nearby cities not exposed to the program. In the short run, the transfer lowered the probability of holding a job by 0.8 percentage points and reduced quarterly earnings by about 2%, driven entirely by declines in part-time work. The implied labor supply elasticities are -0.12 with respect to individual earnings, and effects dissipated immediately after the transfer period. These effects dissipate once payments end, and we detect no differences in employment or total earnings in three years after the transfer period. Overall, the program induces a temporary income effect without longer-run gains in labor-market outcomes.

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1 Introduction

Helping young adults transition into stable, high-quality jobs is a central policy goal in many advanced economies, where prolonged job search and delayed labor market entry have become common. Unconditional cash transfers have drawn growing attention as a way to ease financial constraints during this transition. Recent years have seen a surge of small-scale universal income experiments—often randomized trials among low-income or unemployed adults—in countries such as Finland, Spain, Canada, and the United States, aimed at supporting job search and human capital accumulation (see Appendix Table A.1 for a summary).

However, whether unconditional transfers help young people build stronger employment trajectories or simply postpone labor market attachment remains an open question. Such transfers may improve employment outcomes by relaxing liquidity constraints, enabling longer job search, better matching, or greater investment in skills. But if the additional income primarily finances leisure, they could delay labor force entry or stall skill accumulation. This question lies at the heart of the universal basic income (UBI) debate: advocates argue that unconditional support can facilitate smoother labor market transitions, particularly for young adults at the cusp of economic independence (Van Parijs and Vanderborght, 2017). Yet evidence remains limited: most existing studies are short-term randomized trials on selective populations, offering little insight into long-run or population-wide effects of unconditional income support (Hoynes and Rothstein, 2019).

This paper investigates how unconditional income support affects the labor supply decisions and early employment trajectories of young adults, drawing on evidence from a large-scale quasi-experiment in South Korea. In 2016, the city of Seongnam introduced the Youth Dividend, a universal cash transfer that provided quarterly payments totaling one million KRW (about $850 in 2016, or 19 percent of annual earnings) to all 24-year-olds for one year.[1] Covering an entire age cohort within a city, the program offers a unique opportunity to study the effects of unconditional income support at scale. Because the transfers were directed to individuals rather than households, the setting allows us to isolate the impact of modest, individualized income on young adults' transition to economic independence. The policy's stated goal—to help young adults navigate job search and move into stable employment—speaks directly to an unresolved question in the universal basic income debate: can unconditional transfers at the point of labor market entry shape long-run employment outcomes?

To evaluate the program's effects, we combine confidential administrative data from Statistics Korea linking the Employment Registry and Population Census to construct quar-

[1] This amount is comparable to the Alaska Permanent Fund, one of the first universal basic income programs financed by state oil revenues. Payments in recent years range from $1,000 to $2,000 per year.

terly panels of labor market outcomes. This novel dataset covers the universe of individuals residing in the relevant cities, allowing us to track employment and earnings for all eligible youth over time. To our knowledge, this is the first study to evaluate the Youth Dividend using population-wide data, enabling precise estimates of both short- and long-run impacts at scale.

We compare eligible cohorts of 24-year-olds in Seongnam with those in six nearby cities not exposed to the program, using a stacked difference-in-differences and event-study design following Cengiz et al. (2019). The treatment group comprises the 1992 and 1993 birth cohorts, the first to receive four quarterly payments under the Youth Dividend, while the control group includes individuals of the same age residing in neighboring cities. We focus on youth living with their parents, who account for about 70 percent of the target population and are likely most responsive to the modest transfer amount. The sample is restricted to individuals meeting the policy's residency requirement, with treatment and control status defined by pre-program residence. We follow all individuals—even those who later migrate—to capture their full employment trajectories. The panel spans four quarters before and seven quarters after the program's start for short-run outcomes and extends to three years after the final payment to assess longer-run effects.

Our identification relies on the assumption that, absent the program, labor market trajectories would have evolved similarly across treated and control groups. Event-study estimates confirm parallel pre-trends, supporting the validity of this design. This framework allows us to estimate the causal impact of unconditional income support at the point of labor market entry on both short- and long-run employment outcomes.

We find that the Youth Dividend led to a temporary reduction in labor supply during the transfer period, driven almost entirely by lower participation in part-time jobs. The probability of working in any job with positive earnings declined by 0.8 percentage points from a 32 percent baseline, representing roughly a 2.4 percent reduction in labor supply. The decline was concentrated among part-time and non-daily positions, while regular employment remained largely unaffected. Quarterly earnings fell by about 26,500 KRW—around 2 percent of pre-program earnings—indicating only a modest decline in response to the transfer. Estimated labor-supply elasticities range from –0.12 to –0.41, depending on whether employment probability is defined as working in any job or in part-time jobs only. These estimates are broadly consistent in direction with prior findings. For example, Vivalt et al. (2024) document an elasticity of –0.06 for unconditional cash transfers among low-income U.S. households. Our estimates are larger in magnitude, reflecting that the program in our setting targeted financially dependent young adults. This pattern suggests that younger individuals exhibit more elastic labor-supply responses to temporary income shocks. The higher responsiveness observed here is also consistent with evidence that groups facing fewer fixed commitments display more elastic labor-supply behavior (Blundell and MaCurdy, 1999).

Overall, the results indicate that even modest unconditional transfers can induce short-run reductions in labor supply among young adults. The effects are concentrated in part-time work and appear to reflect substitution away from short-term jobs rather than broader labor-market withdrawal. While these findings align with the program's goal of allowing recipients to temporarily step back from precarious employment to pursue job search or skill investment, it remains unclear whether the additional time was used as intended.

To explore this, we next examine longer-run impacts on employment and earnings two and three years after the final transfer. Employment rates, regular job holding, and annual earnings are statistically indistinguishable between treated and control groups during this period. Although treated individuals were slightly less likely to hold part-time jobs (by 0.3 percentage points) three years after the last payment, this difference did not translate into higher overall employment or earnings. Other indicators of job stability—such as the number of jobs held and total months worked per year—also show negligible differences two and three years after the program ended. The length of the longest job spell in a given year is somewhat shorter among treated individuals in every year since the start of the transfer, but the magnitude—roughly two weeks to one month shorter—is economically insignificant. We also find no discernible effects on migration, indicating that the cash transfer did not induce additional movement across cities. Taken together, these findings suggest that the Youth Dividend's effects were short-lived: while unconditional income temporarily reduced reliance on marginal employment, it did not produce lasting nor meaningful improvements in employability or career progression.

These findings carry important implications for the design of youth-targeted income support programs. While unconditional cash transfers can provide short-term relief and reduce dependence on precarious work, they appear insufficient to improve long-term labor market outcomes. To enhance employability, such programs may need to be complemented by investments in skill development, job placement, or active labor market integration services.

Our study contributes to the growing literature and policy debate on universal basic income (UBI) and unconditional cash transfers (UCTs) by providing new evidence on a youth-targeted program and its long-run labor-market effects.

Much of the prior work evaluates means-tested or conditional income-support programs such as Temporary Assistance for Needy Families (TANF), the Earned Income Tax Credit (EITC), and unemployment insurance (UI), which impose income thresholds or work requirements that distort labor-supply incentives (Ziliak, 2015; Kleven, 2024; Bell et al., 2024). In contrast, unconditional transfers—free of such restrictions—allow researchers to isolate pure income effects. Evidence from developing-country UCTs often shows increases in earnings and labor-force participation through liquidity and entrepreneurship channels (Crosta et al., 2024), but such mechanisms are less relevant in high-income contexts where transfers may instead alter job search intensity or willingness to accept low-quality work.

In high-income settings, findings are mixed. The Alaska Permanent Fund, providing annual dividends of roughly \$1,000–2,000, shows no overall employment effect, likely because increased local demand offsets income reductions (Jones and Marinescu, 2022). Similar null results arise for the Eastern Cherokee demogrant (Akee et al., 2010) and for large-scale pilots such as the Finnish basic-income experiment (Verho et al., 2022; Kangas et al., 2019), the Ontario Basic Income Pilot, and U.S. pilots including Chelsea Eats (Liebman et al., 2022) and Baby's First Years (Stilwell et al., 2024). Historical quasi-UBI and negative-income-tax experiments such as Mincome similarly found small, context-dependent labor-supply effects (Hum and Simpson, 1993; Calnitsky et al., 2019).

The most closely related evidence comes from Vivalt et al. (2024), who conducted a large-scale U.S. randomized experiment providing \$1,000 per month for three years to low-income adults. They find a 3.9-percentage-point ($\simeq$ 5–6 percent) decline in employment and no evidence of improved job quality or other employment outcomes over the study period. Survey and administrative data suggest that younger participants were somewhat more likely to pursue additional education, indicating modest heterogeneity across age groups. Our findings are directionally consistent but show larger short-run labor-supply responses, likely reflecting our focus on financially dependent young adults at labor-market entry. Moreover, by studying a citywide, quasi-universal policy, we address concerns about the limited generalizability of RCTs to universal settings.

Despite this growing evidence, key questions remain about the long-run effects of income support on human-capital accumulation and labor-market attachment, as emphasized by Hoynes and Rothstein (2019). Our study fills this gap using comprehensive administrative data to examine both short- and long-run outcomes. Relative to prior Korean studies that rely on survey data and short-term measures (Yun and Oh, 2021; Yoo et al., 2022), we provide the first population-wide causal evidence on how unconditional income shapes early-career labor dynamics.

This paper proceeds as follows. Section 2 introduces the institutional background, Section 3 describes the data and sample, Section 4 outlines the empirical approach, and Section 5 presents the results. Section 6 concludes.

2 Policy Background

In 2016, Seongnam-si, a suburban city in South Korea, introduced an unconditional cash transfer program targeting young adults, known as the Youth Dividend. The program provided quarterly transfers of 250,000 KRW—amounting to 1 million KRW per year[2]—to all 24-year-old residents who had continuously resided in the city for at least three years. The stated objectives were to provide financial support during the transition to adulthood, facilitate labor market entry, and stimulate local economic activity.

Proposed in mid-2015 and formally announced by Seongnam's mayor that September, the policy was scheduled to launch in 2016 pending approval from the Ministry of Health and Welfare. In December 2015, however, the Ministry publicly rejected the proposal, citing legal and policy concerns. This unexpected rejection created substantial uncertainty about the program's implementation—particularly for the first eligible cohort, individuals born in the first quarter of 1992—whose benefit receipt remained undecided until the launch date. Despite the central government's opposition, Seongnam proceeded with implementation in January 2016 and subsequently filed a legal challenge against the Ministry.

As a result of the initial implementation disruption, individuals born in the first through third quarters of 1992 received only 125,000 KRW in their first quarter of eligibility—half the intended amount. This shortfall was later corrected with a lump-sum payment, ensuring that total annual transfers reached 1 million KRW, as illustrated in Figure A.3. From the 1993 cohort onward, the full quarterly payment of 250,000 KRW was disbursed without interruption. Because the amount distributed to each birth cohort during eligible quarters varied slightly, the average transfer received increased gradually across event quarters, as shown in Figure 2.1. In 2016, during the partial rollout phase, total disbursements amounted to 10.3 billion KRW. Take-up rates were high, reaching 93.9% in 2016 and 96.0% in 2017 (Table 1). According to a government report (Kim et al., 2019), a total of 31.2 billion KRW was distributed to 39,292 individuals over the 2016–2018 period.

Transfers were issued through prepaid electronic cards that could be used only at small local businesses or for utility payments, excluding large retail chains. This design reflected the program's dual objectives of supporting young adults and stimulating local economic activity. Survey data from Seongnam City indicate that the most common uses included groceries and meals, gifts to parents (38%), leisure activities (22.5%), and job-search-related expenses (22.1%) (Kim et al., 2019). Because eligibility was determined by the calendar quarter in which individuals turned 24, the timing of initial receipt varied by birth quarter, introducing additional variation in exposure across cohorts.

[2]At the 2016 average exchange rate of approximately 1,160 KRW per USD, 250,000 KRW corresponds to about $215, and 1 million KRW corresponds to about $860.

3 Data

This study leverages a newly released set of administrative microdata from Statistics Korea that, for the first time, allow researchers to link the Employment Registry with the Population Census at the individual level. This linkage enables us to track the universe of formal-sector workers and residents over an extended time horizon and observe rich employment-related dynamics—including job transitions, earnings, and migration—rarely available outside the Scandinavian countries (Norway, Finland, Sweden, and Denmark).[3]

The Employment Registry contains universe-level employment records for 2015–2023, compiled from mandatory filings with South Korea's four major social insurance programs and the National Tax Service. The data provide detailed job histories for all formal-sector workers, including unique individual identifiers, date of birth, hire and separation dates, employment status, and annual earnings for each job spell. Information on educational attainment, hours worked, and occupation is not available. Contract types are classified into five categories: (i) daily and temporary employment (contracts shorter than one month), (ii) regular employment (contracts longer than one month), (iii) self-employment, (iv) freelance work, and (v) special contracts, which include platform-based or gig work.

Exact start and end dates are observed only for daily, temporary, and regular employment, whereas self-employed, freelance, and special-contract jobs are recorded on an annual basis. For these categories, individuals are flagged as employed if they held such a job at any point during the year, without information on start or end dates. We use this dataset to construct labor market outcomes such as employment, earnings, and job transitions.

The Population Census is a registry-based database of all residents, observed annually each November, and includes information on residential registration and family relationships. We use these data to determine eligibility for the Youth Dividend based on continuous residency within Seongnam for at least three years prior to benefit receipt.

4 Empirical Methodology

We estimate the effects of the Seongnam Youth Dividend on labor market outcomes using a stacked difference-in-differences (DiD) design following Cengiz et al. (2019). Treatment is defined at the birth-quarter-by-city level, with individuals becoming eligible for the transfer in the calendar quarter when they turn 24. Because eligibility timing varies by birth quarter, the correspondence between calendar time and event time differs across cohorts and cities.

[3] To our knowledge, South Korea is the only non-Scandinavian country where individual-level linkage between population and employment registries is available at a national scale. Our team is one of the first to utilize this new dataset since its release by Statistics Korea.

To account for this staggered timing, we construct a balanced panel and realign individuals by their event time—measured in quarters relative to their first transfer receipt—and estimate dynamic treatment effects using a fully interacted event-time specification. This stacked design mitigates the biases of conventional two-way fixed-effects estimators, which can generate negative or non-convex weighting across cohorts and thereby distort treatment effect estimates (Roth et al., 2023). By treating each birth-quarter–region pair as a separate event and stacking them accordingly, we ensure that each treated cohort is compared with an appropriate subset of never-treated individuals. Formally, we estimate the following equation:

$$Y_{itj} = \sum_{k=-4}^{7} \beta_k \times \mathbf{1}\{\text{EventTime}_{it} = k\} \times \text{Treated}_i + \alpha_{jb(i)} + \tau_{tb(i)} + \rho_{jq} + \varepsilon_{itj} \tag{1}$$

where Y_{itj} denotes the outcome of interest for individual i observed in calendar quarter t, residing in region j, and born in birth-year-quarter $b(i)$. Event time k is defined relative to the first quarter in which individual i turns 24—the quarter when the individual becomes eligible for the transfer if residing in *Seongnam-si*. Treated$i$ is an indicator for individuals who resided in *Seongnam-si* before turning age 24 and thus became eligible for the payment. The model includes three sets of fixed effects: (i) birth-year-quarter–by–city fixed effects ($\alpha jb(i)$), which capture cohort–city-specific unobservables; (ii) birth-year-quarter–by–calendar-time fixed effects ($\tau_{tb(i)}$) , which control for cohort-specific time trends; and (iii) city–by–calendar time fixed effects (ρ_{jq}), which capture local business-cycle shocks. Calendar time is measured in quarters, and standard errors are clustered at the city level based on individuals' residence in 2015.

To estimate average effects during and after the transfer period, we collapse the event-time indicators into broader periods and estimate a corresponding difference-in-differences model:

$$Y_{itj} = \beta_d \times \mathbf{1}\{\text{During}\}_t \times \text{Treated}_i + \beta_p \times \mathbf{1}\{\text{Post}\}_t \times \text{Treated}_i + \alpha_{jb(i)} + \tau_{tb(i)} + \rho_{jq} + \varepsilon_{itj} \tag{2}$$

$\mathbf{1}$During$t$ is an indicator for quarters when an individual is age 24—the period of eligibility for the transfer if residing in a treated city—while $\mathbf{1}$Post$t$ indicates the subsequent four quarters when the individual has turned 25 and is no longer eligible. The coefficients of interest, βd and βp, capture average effects on outcome Y_{itj} during and after the transfer period, respectively, relative to the pre-transfer period. We include separate indicators for quarters in which individuals receive the transfer and for quarters following the final payment to assess whether outcomes evolve once the payments end. The specification includes the same fixed effects as the event-study model, and standard errors are clustered at the city level based on 2015 residence.

Identification relies on the standard parallel trends assumption: in the absence of treatment, labor market outcomes for treated and control individuals would have evolved similarly in event time. We assess this assumption by inspecting pre-treatment trends in the event-study coefficients and including multiple leads of treatment. By aligning individuals by policy eligibility rather than calendar time, the stacked design credibly captures dynamic treatment effects and alleviates concerns about negative weighting in staggered adoption settings.

Our primary outcomes are employment and earnings. We first measure employment as an indicator of having any positive income in a given quarter. As described in Section 3, for self-employed, freelance, and special-contract workers—whose job start and end dates are not observed—we code employment as one for all quarters in a year if any positive earnings are reported for that year. We also construct quarterly employment indicators for jobs with observed start and end dates, specifically wage jobs. We further distinguish between regular and part-time employment. Regular employment includes all wage jobs with contract durations longer than one month. Because the data lack information on hours worked, we impute part-time status using monthly earnings at the job level: jobs with average monthly earnings below the full-time minimum-wage threshold (minimum wage $\times$ 209 hours, the statutory full-time hours) are classified as part-time. Finally, we examine earnings. We compute average monthly earnings in a given quarter based on reported start and end dates and annual earnings. Daily or temporary contract jobs, as well as self-employed, freelance, and special-contract work, are excluded because of missing spell information. If an individual holds multiple jobs in a quarter, we aggregate earnings across all jobs.

4.1 Main Analysis Sample

Our main analysis sample consists of individuals born in 1992 and 1993—the first cohorts eligible to receive all four quarterly payments under the Seongnam Youth Dividend program. Individuals born in 1991 were only partially eligible because the program began in January 2016, when eligibility was restricted to residents aged 24. Those born in 1991 qualified for the first payment but aged out before subsequent quarters, resulting in partial exposure. We therefore focus on the 1992 and 1993 cohorts, who were the first to receive the full series of payments.

The treatment group comprises individuals residing in Seongnam-si, while the control group includes individuals in neighboring cities within the same higher-level administrative unit (Gyeonggi Province) that border Seongnam—Hanam, Yongin, Gwacheon, Suwon, Uiwang, and Gwangju.[4]

As a robustness check, we expand the control group to include same-age cohorts in five adjacent districts of Seoul, the capital city. Although Seoul implements its own youth policies—raising the possibility of different time-varying shocks—these bordering districts share similar geographic amenities and household characteristics that may influence youths' labor supply, given Seongnam's proximity to Seoul. We apply the same sample restrictions as in the main analysis: individuals must reside with their parents and remain in the same district prior to treatment. We report results for this extended control group alongside the main event-study estimates. The treated and control cities are shown in dark and light blue, respectively, an

Following the program's stated goal of supporting young adults during the transition to financial independence, we restrict the sample to individuals living with their parents at the time of first cash receipt, identified as those listed as children of the household head in the Population Census. We exclude individuals living independently (about 30% of each birth cohort), for whom small transfers were less likely to be binding, as well as those in group households such as dormitories or orphanages, where living arrangements and labor market behaviors differ systematically.

We construct a balanced panel of individuals observed continuously from four quarters before to seven quarters after their first quarter of eligibility. The program required three years of continuous residency to qualify for benefits. Because our data begin in 2015, we cannot fully verify this criterion for the 1992 cohort, who first received payments in 2016 and for whom only one year of prior residency is observable. For the 1993 cohort, we observe two prior years (2015 and 2016), providing a closer proxy for the three-year rule. To partially enforce the residency requirement, we restrict the 1993 cohort to individuals residing in the same city in both 2015 and 2016. The same restriction is applied to control-group members for consistency.

The number of Seongnam residents in our dataset closely matches the official beneficiary counts reported by Seongnam-si (Table 1), except in 2016 when the counts include the partially eligible 1991 cohort, inflating the number of unique recipients. Based on the Population Census, about 13,000 individuals per cohort (1992–1993 births) lived in Seongnam before the program began, and roughly 10,000 remained continuously resident across 2015–2016—the eligibility window used to define treatment status. Although we can only verify up to two

[4] We primarily use neighboring cities rather than the entire Gyeonggi Province as the control group to ensure comparability. As shown in Appendix Table A.3, average household earnings in the broader province are substantially lower than those in both treated and bordering cities, making the province a less suitable comparison group.

years of prior residency (short of the three-year requirement in the policy), the number of eligible residents identified in our data aligns with official counts. Restricting to continuous residents yields 10,984 (1992) and 12,432 (1993) individuals. Applying additional restrictions—residing in private or group households and inclusion in the balanced event-study window [–4,7]—results in 10,949 and 9,071 individuals, respectively. Further restricting to those co-residing with parents reduces the sample to 8,736 (1992) and 7,710 (1993), which constitute our main treated sample.

Table 2 reports observable characteristics of the main analysis sample, measured in the year prior to individuals' first receipt of the Youth Dividend—that is, the year they turned 23. The sample includes 16,446 treated individuals and 51,892 controls. Individuals in the treated city of *Seongnam* and those in the control cities appear broadly similar in demographic and employment characteristics. Gender composition is nearly identical, with women comprising roughly half of each group. About 32% of treated and 34% of control individuals worked in any job in the year before receiving the benefit. Most employment was in regular or daily jobs, with 31% of treated and 33% of control individuals engaged in these forms of work. By job type, 29% of treated and 31% of control individuals held regular jobs, while 7% in each group worked part-time.

Youths in Seongnam earned on average about 158,800 KRW less per quarter than those in the control cities. However, household income levels are comparable across groups: average annual household income is 53.5 million KRW among treated and 49.7 million KRW among control individuals. Per capita household income is also similar, with treated individuals from households earning about 820,000 KRW ($\simeq$\$725 in 2015 USD) more than controls. The extended control group—including neighboring districts—shows broadly similar patterns (Table A.2). Earnings and household income gaps narrow further when using this enlarged control group. The Youth Dividend amounted to 250,000 KRW per quarter for one year (1 million KRW in total), equivalent to roughly 19.1% of treated individuals' baseline total annual earnings - 524.7 million KRW.

5 Impacts on Youth Labor Market Outcomes

5.1 Short-Run Effects on Labor Supply

We begin by examining the short-run effects of the Youth Dividend on treated individuals' labor supply. Event study results are presented in Figure 2.2 and Figure 3, with corresponding difference-in-differences estimates reported in Panel A of Table 3. Table 3 summarizes average effects over two key periods: the four quarters in which individuals received the transfer ($During\ Transfer$) and the four quarters following the final payment ($Post\ Transfer$).

We find that the probability of working temporarily declines during the cash receipt periods and rebounds once payments end, as shown in Figure 2.2. Event study estimates show negative and statistically significant effects only during the receipt periods. Because event time is aligned by each cohort's first quarter of benefit eligibility (the quarter they turn 24), these patterns are unlikely to reflect seasonal or age-related labor market changes. Notably, while most new job entries in Korea occur in the first or third calendar quarters, the observed decline in earnings coincides precisely with benefit receipt periods rather than with calendar time. Moreover, as the average transfer amount gradually increased over the disbursement period due to administrative processes (Figure 2.1), we observe a corresponding gradual decline in labor supply, mirroring this increase in transfer size.

This pattern is consistent across alternative control group definitions—both when compared to same-birth-cohort individuals in neighboring cities within Gyeonggi Province (our main control group) and when we additionally include same-birth-cohort individuals residing in adjacent districts of Seoul. Averaged across the four quarters of cash receipt, the probability of working in any job with positive earnings declines by 0.78 percentage point from a baseline of 32%, representing a 2.4% reduction in labor supply (Column 1 of Panel A in Table 3). This outcome measure includes self-employed, freelance, and special-contract workers, for whom start and end dates are not reported; we therefore assume these individuals are employed throughout the year. Redefining employment as a binary indicator for working in jobs that report both start and end dates—including both regular and daily jobs—yields a similar temporary decline in employment probability, as shown in Figure 3a. The corresponding difference-in-differences estimate indicates a 0.75-percentage-point reduction from a baseline of 31% (Column 2 of Panel A in Table 3).

Figure 3b shows that the decline is primarily driven by individuals in non-daily positions, or regular jobs that involve contract terms longer than one month, and is concentrated among part-time workers (Figure 3d). Full-time and part-time jobs are distinguished based on earnings within each job. Column 4 of Panel A in Table 3 indicates a 0.6-percentage-point decline in the probability of holding a part-time job—similar in magnitude to the overall reduction reported in Columns 1–3—suggesting that the temporary drop in labor supply is mainly attributable to reduced participation in part-time work. Because regular employment encompasses both full- and part-time positions, this pattern implies that the transfer primarily reduced reliance on transitory, short-term employment.

Consistent with these findings, Figure 3c shows a corresponding decline in quarterly earnings during the transfer period. Average earnings fall by 26,500 KRW per quarter relative to the pre-treatment mean of 1.24 million KRW (Column 5 of Panel A in Table 3), corresponding to a 2% reduction. Given that recipients received an average transfer of 250,000 KRW per quarter, the associated earnings reduction represents only about 10% of the transfer amount. Thus, most of the transfer did not translate into foregone earnings, suggesting a limited short-run labor supply response.

To capture labor supply responses more precisely, we exploit variation in the amount of cash received across quarters. As described in Section 2, the 1992 birth cohort experienced uneven quarterly payments due to administrative delays: early cohorts received smaller transfers in the first three quarters and a lump-sum payment in the final quarter (Figure A.3). In contrast, the 1993 cohort received a uniform 250,000 KRW each quarter. This within-event time variation allows us to estimate the response to the actual transfer amount using the following specification:

$$Y_{itj} = \text{Transfer Received}_{itj} \cdot \beta + \alpha_{jb(i)} + \tau_{tb(i)} + \rho_{jq} + \varepsilon_{itj}$$

where $\text{TransferReceived}_{itj}$ denotes the amount (in 1 million KRW) received in quarter t by individual i in city j, set to zero for all control individuals and for treated individuals after the transfer period. The coefficient β measures the change in the outcome associated with an additional 1 million KRW received in a given quarter.We scale the variable in 1 million KRW units—the total annual transfer amount—for ease of interpretation, allowing direct comparison with the binary treatment estimates in Panel A. Results are reported in Panel B of Table 3.

An additional 1 million KRW in transfers reduces the probability of working by about 1.6 percentage points, with the effect concentrated among part-time workers whose employment probability falls by roughly 2 percentage points. These estimates are broadly consistent with the binary treatment results in Panel A—showing similar magnitudes and directions. Quarterly earnings also decline with higher transfers, though the estimates are less precise: a 1 million KRW increase in transfers lowers earnings by about 72,000 KRW. Using the actual cash amount received per quarter yields slightly smaller but statistically similar effects, reinforcing the main findings.

To interpret these magnitudes, we compute the implied labor-supply elasticities with respect to dividends using the probability of working in regular and daily jobs (Column 2 of Table 3), which includes jobs with accurate start- and end-date information. Treated individuals reduced their employment probability by 0.75 percentage points per quarter from a 31% baseline. Given that recipients received 250,000 KRW per quarter—about 20.2% of baseline quarterly earnings—this corresponds to an elasticity of –0.12, meaning a 10%

increase in income leads to a 1.2 percentage-point decline in employment probability. Part-time labor supply is more elastic: a 0.58 percentage-point reduction in part-time employment from a 7% baseline implies an elasticity of –0.41, where a 10% increase in earnings corresponds to a 4.1 percentage-point decline in part-time employment. Using household income as the denominator yields a larger elasticity, as the annual transfer equals 6.5% of baseline household income (15.7 million KRW), implying an elasticity of –0.37.

These values are broadly consistent with prior studies. Vivalt et al. (2024) find that an unconditional transfer of $12,000 over three years (about 56% of annual baseline income) reduced employment by 2 percentage points from a 58% baseline, implying an elasticity of –0.06. Our estimates are larger in magnitude, reflecting the fact that the program in our setting targeted financially dependent young adults. This suggests that younger individuals exhibit more elastic labor-supply responses to temporary income shocks. The higher responsiveness in our setting is consistent with evidence that groups facing fewer fixed commitments exhibit more elastic labor-supply responses to temporary income shocks (Blundell and MaCurdy, 1999).

Overall, the results indicate that even modest, unconditional transfers can induce short-run reductions in labor supply among young adults. The effects are concentrated in part-time work and are consistent with the program's goal of enabling recipients to temporarily step back from low-quality, short-term jobs. By easing short-term income pressures during the transition to adulthood, the policy may have facilitated more forward-looking decisions, such as job search or skill investment. The next section examines whether these short-run adjustments translate into longer-term improvements in employment outcomes.

5.2 Long-Run Effects on Employment Outcomes

Does the temporary reduction in part-time work observed during the transfer period translate into improved employment outcomes in the longer run? One of the main policy objectives of the Youth Dividend was to enable young adults to invest in career-enhancing activities—such as job search, skills training, or further education—by reducing their reliance on low-quality, short-term jobs and ultimately facilitating transitions into higher-quality employment.

However, existing evidence casts doubt on the strength of this mechanism. A city government survey of Youth Dividend recipients indicates that most of the transfer was spent on groceries and dining out (Kim et al., 2019). Likewise, Vivalt et al. (2024) find that unconditional cash transfers have no meaningful effects on employment quality; their confidence intervals rule out even modest improvements in job search intensity or time spent on self-development. Although they report some suggestive evidence of increased educational enrollment among young recipients, their sample of young adults is relatively limited compared to ours. Whether unconditional cash transfers to financially dependent young adults can improve long-run labor market outcomes therefore remains an open question.

If such transfers do induce forward-looking behavioral responses, the temporary reduction in part-time work documented in Section 5.1 could reflect time reallocated toward career development. Although we lack survey-based time-use data during the transfer period, we exploit our administrative panel to examine employment outcomes up to three years after the final payment. To our knowledge, this is the first study to estimate the long-run impacts of the Youth Dividend. Specifically, we estimate a difference-in-differences model of the form:

$$\begin{aligned} Y_{itj} = {} & \beta_1 \mathbf{1}\{\text{During}\}_t \times \text{Treated}_i + \beta_2 \mathbf{1}\{\text{After 1yr}\}_t \times \text{Treated}_i \\ & + \beta_3 \mathbf{1}\{\text{After 2yr}\}_t \times \text{Treated}_i + \beta_4 \mathbf{1}\{\text{After 3yr}\}_t \times \text{Treated}_i + \alpha_{jb(i)} + \tau_{tb(i)} + \rho_{jq} + \varepsilon_{itj}, \end{aligned} \tag{3}$$

where the pre-period is defined as the four quarters preceding the start of transfers. $\mathbf{1}$During$t$ indicates the four quarters during which recipients receive the transfer. $\mathbf{1}$After 1yr$t$ corresponds to the four quarters in the first year following the final payment, while $\mathbf{1}$After 2yr$t$ and $\mathbf{1}$After 3yr$t$ represent the four quarters in the second and third years after the final payment, respectively.[5]

We include the same set of fixed effects as in the short-run analysis: birth-year–quarter–by–city fixed effects ($\alpha_{jb(i)}$), birth-year–quarter–by–calendar-time fixed effects ($\tau_{tb(i)}$), and city–by–calendar-time fixed effects (ρ_{jq}). We analyze the same outcomes as in the main analysis, reported in Panel C of Table 3. We report only the coefficients on β_3 and β_4 in Table **??** to evaluate longer-run impacts since β_1 and β_2 are identical estimates from Panel A of Table 3. Finally, we complement these estimates with event-study results extending up to three years post-payment (Appendix Figure A.1 and Figure A.2).

We find no evidence of improved employment probabilities (Columns 1–3 of Panel C in Table 3) nor higher earnings (Column 5 of Panel C). The only notable result is a 0.3-percentage-point decline in part-time employment—equivalent to a 4.3% reduction relative to the treat group's pre-period average. Although treated individuals were less likely to hold part-time jobs three years after the last payment, this adjustment did not translate into higher overall employment or earnings.

To further examine employment-related outcomes, we analyze quarterly indicators that capture job stability and labor-market attachment: (i) the number of jobs held during a given quarter, (ii) the total number of months worked across all jobs within the quarter (capped

[5] We analyze outcomes up to three years after the final payment because later periods (2020–2021) coincide with the onset of COVID-19 and numerous concurrent policy interventions, complicating causal interpretation.

at three months), and (iii) the longest job spell observed within the quarter (Table 4). Each outcome is constructed from job records that report both start and end dates, including regular, temporary, and daily positions. We estimate the same specification as in Panel C of Table 3.

During the transfer period, treated individuals exhibit a statistically significant decline in both the number of jobs held and total months worked per quarter, consistent with the temporary reduction in employment probability shown in the main results (Column 1 and 2). This pattern suggests that recipients were less likely to engage in short-term or multiple job spells while receiving the transfer. In the one to three years following the final payment, however, the estimates for both outcomes are small and statistically insignificant, indicating no differential trends in job switching or overall labor supply between treated and control groups after the transfer period.

Column 3 of Table 4 shows that the longest job spell within a quarter is modestly shorter for treated individuals—by roughly 0.25 to 0.4 months (about one to two weeks)—during and shortly after the transfer. This difference most plausibly reflects delayed entry into stable full-time employment rather than differences in ongoing job retention, since we find no significant changes in the number of jobs (indicating similar rates of job switching) or in total months worked (reflecting comparable overall labor supply). The magnitude and pattern of estimates further support this interpretation: the initial gap in average tenure—about one week shorter among treated individuals during the transfer period—appears to persist into subsequent years as a carryover effect. For instance, the year-one estimate of –0.37 months implies that roughly 0.28 months (about one week) stem from the initial delay, with only an additional –0.09 months (around three days) widening thereafter. The year-two estimate remains similar (–0.37 months), suggesting no further divergence, and the year-three estimate of –0.42 months corresponds to just an extra –0.05 months (about one day). These small cumulative differences indicate that the observed tenure gap primarily arises from slightly later entry into full-time jobs during the transfer period, with no meaningful subsequent divergence in employment stability or job-switching behavior.

We also test whether the Youth Dividend affected migration behavior. Using annual residential address records from the Census, we construct an indicator equal to one if an individual moved out of their baseline city in a given year. As shown in Figure **??**, treated individuals exhibit a modest decline in out-migration relative to the control group, suggesting that the program may have slightly discouraged migration; however, the estimates are statistically insignificant, indicating no robust evidence of an effect on residential mobility.[6]

[6] We also compare the characteristics of migrants between treated and control groups who moved during the benefit period and those who moved two years after the final payment, estimating difference-in-differences

Taken together, these results indicate that the Youth Dividend had no discernible longer-run effects on employment stability, job quality, or job search–related behaviors such as migration.

6 Conclusion

This paper evaluates the effects of the Seongnam Youth Dividend—a universal cash transfer program providing quarterly payments to all 24-year-olds—on labor market outcomes. Using linked administrative data, we track individuals for up to three years after the final payment to examine how the program affected employment and earnings during and after the transfer period. Our findings show that the Youth Dividend had limited effects beyond the transfer period. We observe only a modest, short-lived decline in part-time employment, with no discernible impacts on earnings, overall employment, or job stability. The temporary reduction in part-time work during the transfer period did not translate into sustained improvements in labor market outcomes. These results are consistent with prior evidence that unconditional cash transfers may offer short-term flexibility without generating lasting gains in employment quality or income (Vivalt et al., 2024).

By following participants over a three-year horizon, this study provides one of the few pieces of evidence on the longer-run labor market effects of unconditional youth cash transfers. The results suggest that such programs, while potentially easing short-term financial constraints, have limited capacity to alter employment trajectories once transfers end. The lack of measurable improvements in employment or earnings highlights the constraints of one-time or short-duration unconditional transfers as stand-alone policy tools.

Future research can build on these findings along several dimensions. First, extending the analysis beyond three years would offer valuable insight into whether effects emerge later in adulthood—particularly as individuals complete transitions into stable employment or family formation. Doing so would require incorporating data from 2020–2021, a period that coincides with the onset of COVID-19 and overlapping policy responses; careful adjustment for these shocks could reveal longer-term impacts that our pre-pandemic window cannot capture. Second, an important avenue for future work is to examine the policy's local economic effects, which were a key stated objective of the Youth Dividend. Because the transfers were distributed as a local currency redeemable only at businesses within Seongnam-si,

models for employment status, per capital household income, total annual earnings at the time of migration. and the ratio of female among migrants (Table A.4). The results show no significant differences across treated and control group among those who migrate in during and after the transfer, implying that while the overall rate of migration may have declined slightly, the composition of migrants remained similar across groups.

evaluating potential spillover effects on local consumption, employment, or business activity would provide a more comprehensive assessment of the program's overall welfare impact.

Overall, our results underscore the limited capacity of temporary, universal cash transfers to generate durable labor market improvements among young adults. Policymakers considering similar programs may consider pairing them with targeted or sustained interventions—such as training, job placement services, or continued income support—to enhance their effectiveness and better address the challenges facing financially dependent youth during early career transitions.

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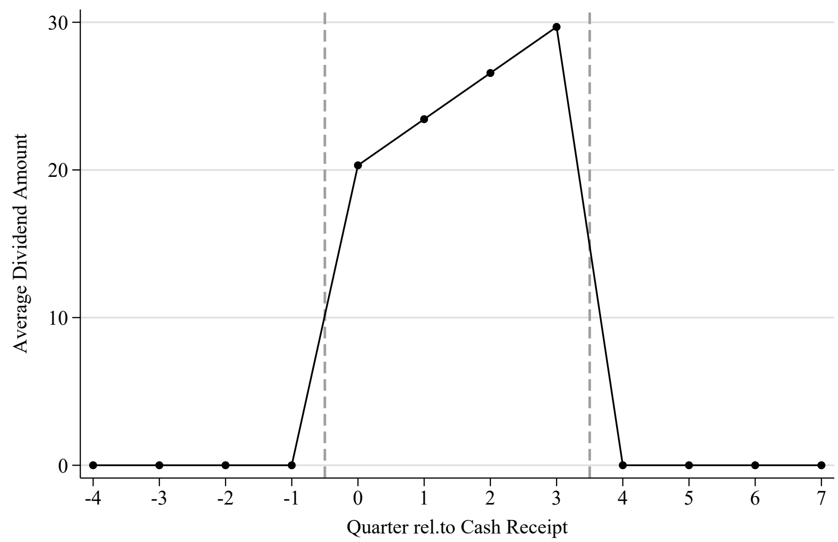
7 Figure&Table

Figure 1: Map of Treated and Control Cities in Gyeonggi Province

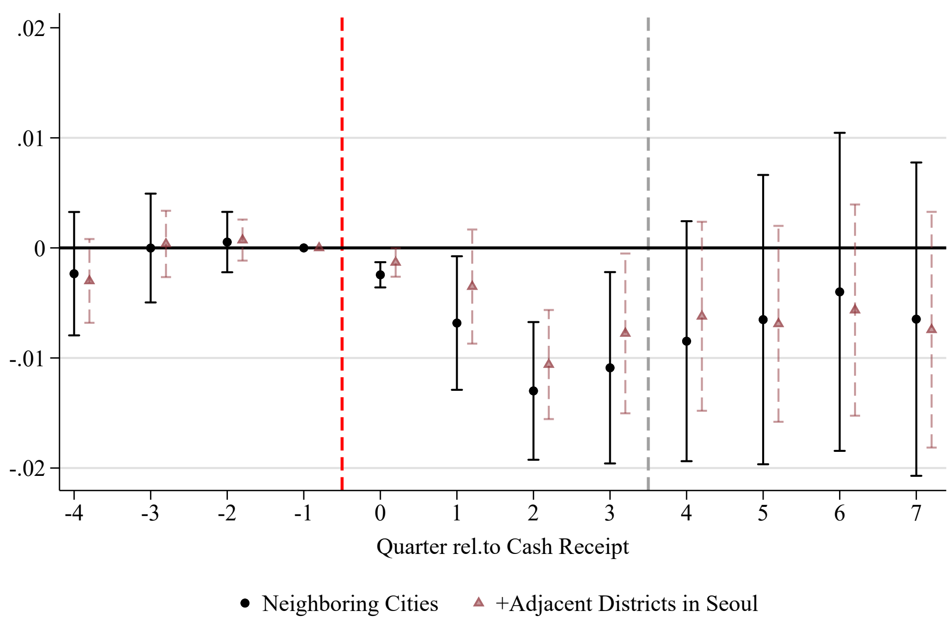
Notes: This figure shows a map of Gyeonggi Province, highlighting the cities used in the main analysis. The darkest-shaded area represents Seongnam-si, the treated city that implemented the Youth Dividend program. The surrounding labeled cities with lighter shading constitute the control group in the main analysis. In addition, the dotted areas indicate an expanded control group that includes neighboring districts within the Seoul Metropolitan Area, which is used for robustness checks.

Figure 2.1: Average Dividend Amount



Notes: This figure shows the average dividend amount received by the treated group over time.

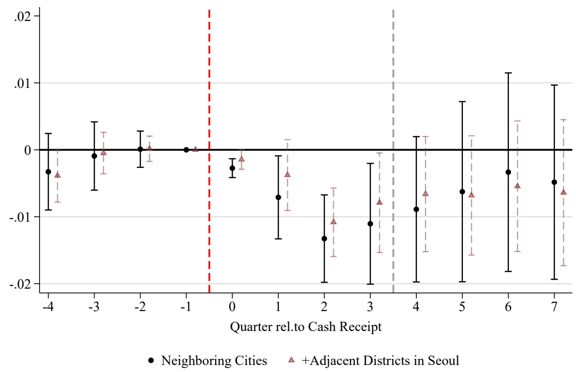
Figure 2.2: Effects of Cash Transfer on Employment



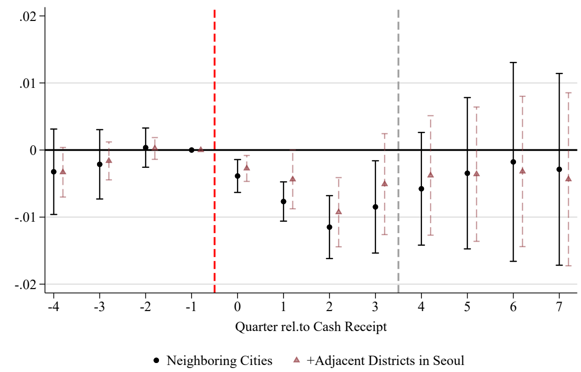
Notes: The figures present dynamic difference-in-differences estimates of the impact of the Youth Dividend on employment(including all employment status). The analysis sample includes individuals from the 1992 and 1993 birth cohorts residing with their parents in Seongnam or control cities, restricted to those continuously observed in the panel. The specification includes birth-year-quarter × city, birth-year-quarter × calendar time, and city × quarter fixed effects. Each dot represents the interaction coefficient between treatment status and event time, and capped lines indicate 95% confidence intervals. Standard errors are clustered at the city level.

Figure 3: Effects of Cash Transfer on Labor Supply

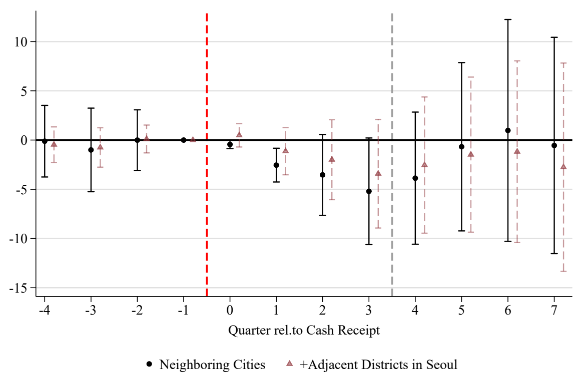
(a) Any working (Regular+Daily+Temporary)



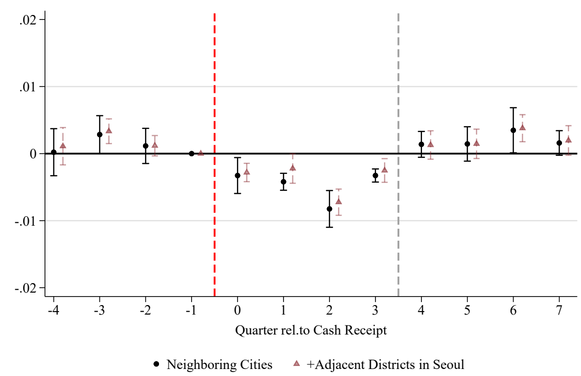
(b) Regular job



(c) Quarterly earnings(Regular)



(d) Parttime Job



Notes: The figures present dynamic difference-in-differences estimates of the impact of the Youth Dividend on regular and daily employment (Panel (a)), regular employment (Panel (b)), quarterly earnings of regular workers (Panel (c)) and part-time employment (Panel (d)). The sample and fixed effects are the same as in Figure 2.2. Each dot represents the interaction coefficient between treatment status and event time, and capped lines indicate 95% confidence intervals. Standard errors are clustered at the city level.

Table 1: Annual Youth Dividend Participation Rates in Seongnam (2016–2018)

Year	Recipients	Take up Rate	Total Payment Amount (1 million KRW)
2016	18,324	93.90%	10,360
2017	10,603	96.00%	10,546
2018	10,365	93.00%	10,283
Total	39,292	94.20%	31,189

Notes: This table reports the number of eligible individuals and recipients of the Seongnam Youth Dividend program between 2016 and 2018. The information is from the government report

Table 2: Descriptive Statistics of the Main Analysis Sample

	Treated (Seongnam)	Control (Other Cities)	Difference
Female	0.51	0.52	-0.01
	(0.50)	(0.50)	(0.01)
Ever worked	0.32	0.34	-0.02
	(0.47)	(0.47)	(0.01)
Ever worked (Regular+Daily+Temporary)	0.31	0.33	-0.02
	(0.46)	(0.47)	(0.01)
Regular job	0.29	0.31	-0.02
	(0.45)	(0.46)	(0.01)
Part-time job	0.07	0.07	0.00
	(0.25)	(0.25)	(0.00)
Unit: 10,000 KRW			
Quarterly earning (Regular)	124.01	139.89	-15.88**
	(229.59)	(246.90)	(4.96)
Household income	5353.63	4965.37	388.26
	(10511.64)	(6038.68)	(206.39)
Per capita household income	1572.40	1490.50	81.91
	(2766.54)	(1662.11)	(50.32)
N Individuals	16,446	51,892	

Notes: This table reports means and standard deviations (in parentheses) for individuals in the treated and control cities. Column (3) shows the difference in means and standard error of difference. Earnings variables are reported in units of 10,000 KRW. *** indicates statistical significance at the 1% level.

Table 3: Effect of Cash Transfer on Earnings and Employment Outcomes

	(1) Ever Worked	(2) Ever Worked (Regular+Daily)	(3) Regular Job	(4) Part-time Job	(5) Quarterly Earnings (10k KRW)
Panel A. Baseline Difference-in-Differences Estimates					
Treated × 1(During Transfer year)	-0.0078***	-0.0075***	-0.0066***	-0.0058***	-2.6520**
	(0.0015)	(0.0016)	(0.0017)	(0.0006)	(0.9540)
Treated × 1(After 1year)	-0.0059	-0.0048	-0.0022	0.0009	-0.7490
	(0.0046)	(0.0048)	(0.0046)	(0.0009)	(3.328)
Constant	Yes	Yes	Yes	Yes	Yes
City × Birth Year-Quarter FE	Yes	Yes	Yes	Yes	Yes
City × Calendar Time FE	Yes	Yes	Yes	Yes	Yes
Calendar Time × Birth Year-Quarter FE	Yes	Yes	Yes	Yes	Yes
Observations	820,056	820,056	820,056	820,056	820,056
Panel B. Effects of Transfer Amount					
Transfer amount	-0.0163***	-0.0169***	-0.0183***	-0.0203***	-7.1640*
	(0.0018)	(0.0015)	(0.0044)	(0.0022)	(3.0900)
Constant	Yes	Yes	Yes	Yes	Yes
City × Birth Year-Quarter FE	Yes	Yes	Yes	Yes	Yes
City × Calendar Time FE	Yes	Yes	Yes	Yes	Yes
Calendar Time × Birth Year-Quarter FE	Yes	Yes	Yes	Yes	Yes
Observations	820,056	820,056	820,056	820,056	820,056
Panel C. Long-term Effects					
Treated × 1(After 2year)	0.0016	0.0041	0.0055	-0.0017	6.1530
	(0.0068)	(0.0071)	(0.0075)	(0.0009)	(5.7980)
Treated × 1(After 3year)	0.0018	0.0049	0.0055	-0.0029**	7.8700
	(0.0089)	(0.0093)	(0.0102)	(0.0009)	(8.8330)
Constant	Yes	Yes	Yes	Yes	Yes
City × Birth Year-Quarter FE	Yes	Yes	Yes	Yes	Yes
City × Calendar Time FE	Yes	Yes	Yes	Yes	Yes
Calendar Time × Birth Year-Quarter FE	Yes	Yes	Yes	Yes	Yes
Observations	1,359,974	1,359,974	1,359,974	1,359,974	1,359,974

Notes: This table reports the estimated effects of the Youth Dividend on labor market outcomes. The dependent variables are listed in column headers. ”During Transfer” refers to the four quarters in which individuals received the benefit; ”After Transfer” refers to the four quarters following the last payment. All models include city-by-birth cohort fixed effects, city-by-quarter fixed effects, and calendar-by-birth cohort fixed effects. Robust standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

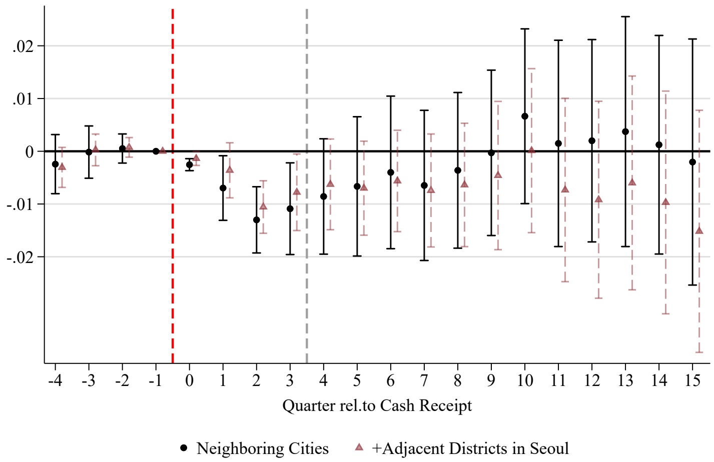
Table 4: Effect of Cash Transfer on Medium and Long Run Outcomes

	(1) Number of Jobs	(2) Total Months Worked	(3) Length of Longest Job Spell (month)
Treated × 1(During Transfer year)	-0.0109***	-0.0189**	-0.281***
	(0.0022)	(0.0065)	(0.0490)
Treated × 1(After 1 year)	-0.0047	-0.0068	-0.372***
	(0.0047)	(0.0141)	(0.0259)
Treated × 1(After 2 years)	0.0090	0.0216	-0.371***
	(0.0079)	(0.0190)	(0.0431)
Treated × 1(After 3 years)	0.0104	0.0263	-0.420**
	(0.0099)	(0.0243)	(0.1220)
Constant	Yes	Yes	Yes
City × Birth Year-Quarter FE	Yes	Yes	Yes
City × Calendar Time FE	Yes	Yes	Yes
Calendar Time × Birth Year-Quarter FE	Yes	Yes	Yes
Observations	1,359,974	1,359,974	1,359,974

Notes: This table reports the estimated effects of the Youth Dividend on annual income and employment outcomes three and six years after program implementation. The dependent variables are listed in the column headers. Estimates in the top panel correspond to outcomes three years after receipt of the benefit, and those in the bottom panel correspond to outcomes six years after receipt. All models include birth year–by–city and birth year–by–year fixed effects. Robust standard errors are reported in parentheses. The values in the Control Post Avg row represent the mean of the dependent variable for the control group in the respective period (three or six years after receipt), and the % Rel. to Control Mean row expresses the magnitude of the reported coefficient relative to that mean. *** p<0.01, ** p<0.05, * p<0.1.

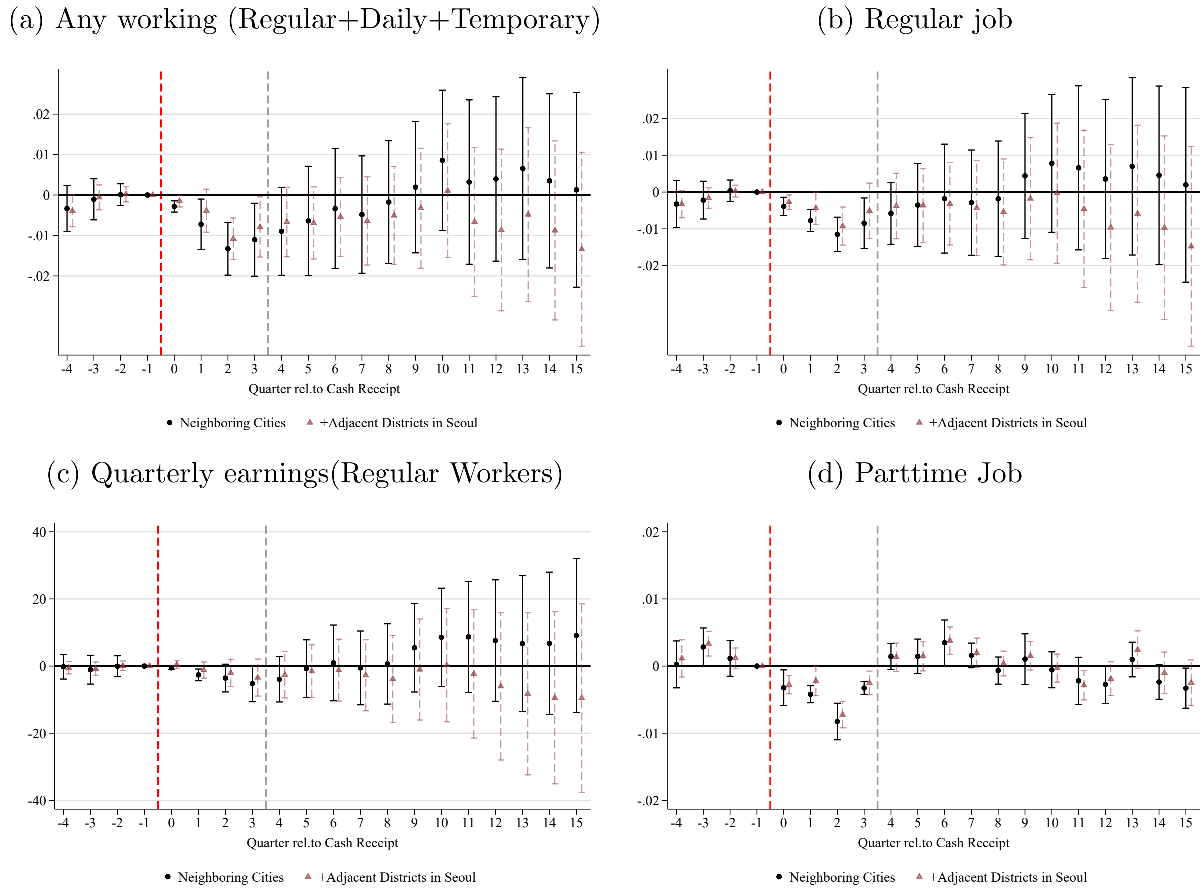
Appendix

Figure A.1: Effects of Cash Transfer on Employment



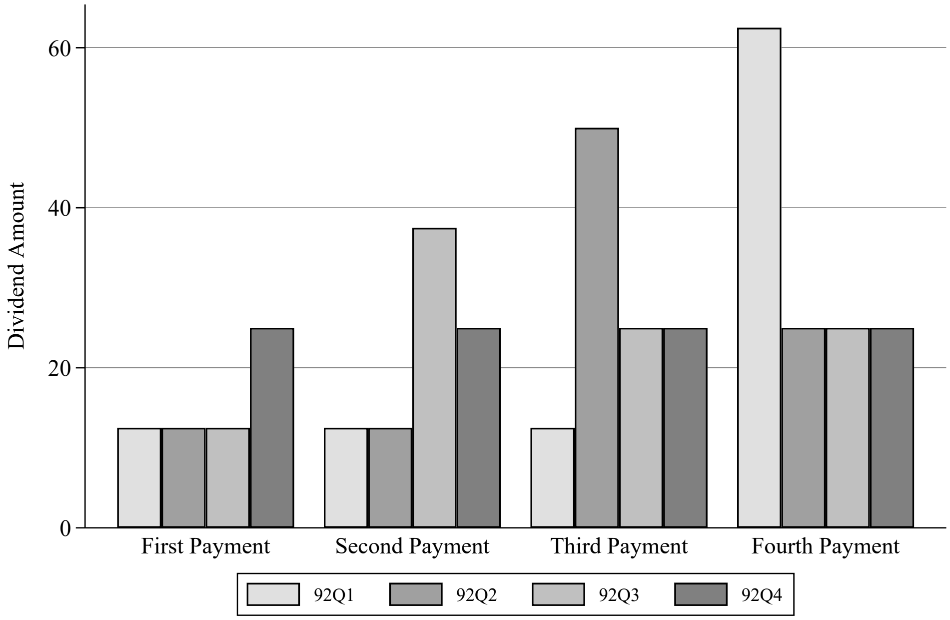
Notes: The figures present dynamic difference-in-differences estimates of the impact of the Youth Dividend on employment(including all employment status). The analysis sample includes individuals from the 1992 and 1993 birth cohorts residing with their parents in Seongnam or control cities, restricted to those continuously observed in the panel. The specification includes birth-year-quarter × region, birth-year-quarter × calendar time, and region × quarter fixed effects. Each dot represents the interaction coefficient between treatment status and event time, and capped lines indicate 95% confidence intervals. Standard errors are clustered at the city level.

Figure A.2: Effects of Cash Transfer on Labor Supply



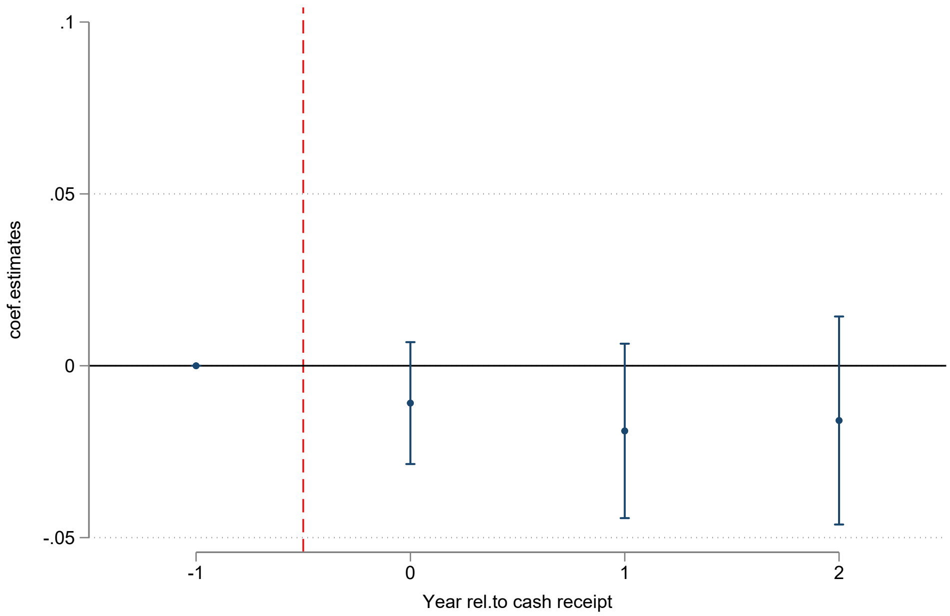
Notes: The figures present dynamic difference-in-differences estimates of the impact of the Youth Dividend on employment(regular, daily job) (Panel (a)), regular job (Panel (b)), quarterly earnings of regular workers (Panel (c)) and part-time employment (Panel (d)). The sample and fixed effects are the same as in Figure 2.2. Each dot represents the interaction coefficient between treatment status and event time, and capped lines indicate 95% confidence intervals. Standard errors are clustered at the city level.

Figure A.3: Quarterly Transfer Amounts by Birth Quarter (1992 Cohort)



Notes: This table reports the quarterly payment amounts received by individuals born in each quarter of 1992 under the Seongnam Youth Dividend program. Due to administrative delays and legal uncertainty during the program's initial rollout, individuals born in earlier quarters received partial payments in early 2016, followed by lump-sum reimbursements in December 2016 to ensure the total annual transfer reached 1 million KRW. From 2017 onward, quarterly payments of 250,000 KRW were disbursed as originally planned. Individuals in the 1993 cohort received regular quarterly payments of 250,000 KRW without disruption. The information is based on administrative records from Seongnam City.

Figure A.4: Effects of Cash Transfer on Migration



Notes: This event study figure presents dynamic difference-in-differences estimates of the impact of the Youth Dividend on the probability of out-migration. The analysis sample includes individuals from the 1992 and 1993 birth cohorts residing with their parents in Seongnam or control cities, observed annually through 2023. The outcome is an indicator equal to one if an individual moved out of their baseline city in a given year. The specification includes birth-year-quarter $\times$ city, birth-year-quarter $\times$ calendar time, and city $\times$ year fixed effects. Each dot represents the interaction coefficient between treatment status and event year, and capped lines indicate 95% confidence intervals. Standard errors are clustered at the city level.

Table A.1: Comparative Summary of Recent UBI/UCT Initiatives in High-Income Countries (Simplified)

Country	Program	Type	Amount/Value	Eligibility Restriction
Finland	Basic Income Experiment	RCT	€560 per month	Ages 25–58 receiving unemployment benefits
Germany	Pilotprojekt Grundeinkommen	RCT	€1,200 per month	Ages 21–40 with median income; single-person households
U.S. (Stockton, CA)	Stockton Economic Empowerment Demonstration (SEED)	RCT	$500 per month	None
U.S. (Two States)	Guaranteed Income RCT (ORUS)	RCT	$12,000 per year	Ages 21–40 with low income
Canada (Ontario)	Basic Income Pilot (OBIP)	RCT	Up to CAD 16,989 per year, reduced by $0.50 for every dollar earned	Ages 18–64 with low income

Table A.2: Descriptive Statistics of the Robustness Test Sample

	Treated (Seongnam)	Control (Other Cities)	Difference
Female	0.51	0.52	-0.01**
	(0.50)	(0.50)	(0.01)
Ever worked	0.32	0.31	0.01
	(0.47)	(0.46)	(0.02)
Ever worked (Regular+Daily)	0.31	0.31	0.01
	(0.46)	(0.46)	(0.02)
Regular job	0.29	0.28	0.01
	(0.45)	(0.45)	(0.02)
Part-time job	0.07	0.07	0.00
	(0.25)	(0.25)	(0.00)
Unit: 10,000 KRW			
Quarterly earning (Regular)	124.01	125.08	-1.073
	(229.59)	(236.84)	(7.56)
Household income	5353.63	5570.52	-216.89
	(10511.64)	(9124.85)	(480.30)
Per capita household income	1572.40	1632.89	-60.48
	(2766.54)	(2405.34)	(120.34)
N Individuals	16,446	100,562	

Notes: This table reports means and standard deviations (in parentheses) for individuals in the treated and control cities. Column (3) shows the difference in means and standard error of difference. Earnings variables are reported in units of 10,000 KRW. *** indicates statistical significance at the 1% level.

Table A.3: Descriptive Statistics of All Sample

	(Treated) Seongnam	(Main Control) Neighboring Gyeonggi Cities	Other Gyeonggi Cities
Female	0.51	0.52	0.51
	(0.50)	(0.50)	(0.50)
Ever worked	0.32	0.34	0.36
	(0.47)	(0.47)	(0.48)
Ever worked (Regular+Daily)	0.31	0.33	0.36
	(0.46)	(0.47)	(0.48)
Regular job	0.29	0.31	0.33
	(0.45)	(0.46)	(0.47)
Part-time job	0.07	0.07	0.07
	(0.25)	(0.25)	(0.26)
Unit: 10,000 KRW			
Quarterly earning (Regular)	124.01	139.89	150.81
	(229.59)	(246.90)	(248.97)
Household income	5353.63	4965.37	4136.87
	(10511.64)	(6038.68)	(4952.56)
Per capita household income	1572.40	1490.50	1287.84
	(2766.54)	(1662.11)	(1360.27)
N Individuals	16,446	51,892	159,381

Notes: This table reports means and standard deviations (in parentheses) for individuals in the treated and control cities. Column (1) reports estimates for the treated city, Seongnam. Column (2) corresponds to the main analysis control group, defined as neighboring cities within Gyeonggi Province. Column (3) represents cities in Gyeonggi Province excluding the neighboring cities. Earnings variables are reported in units of 10,000 KRW. *** indicates statistical significance at the 1% level.

Table A.4: Comparison of Migrant Characteristics During and After the Transfer Period

	(1) Ever Worked	(2) Regular Job	(3) Total Annual Income	(4) Per Capita Household Income	(5) Female
Treated × 1(Migrated 2 Years After Transfer)	0.0084	0.0175	65.6924***	16.5906	-0.0124
	(0.0119)	(0.0121)	(11.5385)	(35.1774)	(0.0121)
Observations	9,992	9,992	9,992	9,992	9,992
Birth Year × City FE	Yes	Yes	Yes	Yes	Yes
Birth Year × Year FE	Yes	Yes	Yes	Yes	Yes

Notes: This table reports difference-in-differences estimates comparing migrants from treated and control groups who moved during the transfer period and those who moved two years after the final payment. Each column corresponds to a separate regression using the specified outcome as the dependent variable. All models include birth-year–by–city and birth-year–by–calendar-year fixed effects. Robust standard errors are reported in parentheses. *** indicates statistical significance at 1% level, ** indicates 5% level, and * indicates 10% level.